

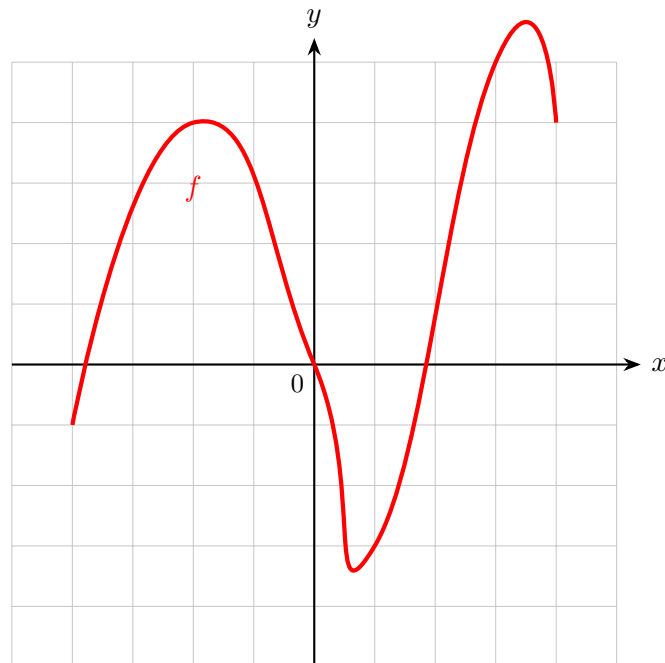
Quiz # 4

Math 118
Instructor: _____
Fall 2025

Names: _____

To receive full credit you must show all work and provide clear, complete explanations. Use of books, notes, calculators, etc. is not permitted.

1. (6 points) Use the given graphs of f and g to evaluate the expressions.



(a) $(f \circ g)(1)$

(b) $\left(\frac{f}{g}\right)(-1)$

(c) $(f \cdot g)(3)$

2. (4 points) Let $f(x) = \frac{1+x}{x^2}$ and $g(x) = x^3$

(a) Find $(f \circ g)(x)$.

(b) Find $(g \circ f)(-\frac{1}{2})$

3. (5 points) Verify that the two functions:

$$f(x) = \frac{3}{2x-1}, \quad g(x) = \frac{3+x}{2x}$$

are inverses of each other.

4. (5 points) Find the inverse of:

$$y = \frac{3x}{x - 2}$$